Bond Valuation (Chapter 7) - Complete Notes

Corporate Finance

Fundamental Time Value of Money Formulas

Core TVM Equations

$$\text{Future Value (Single Sum)}: \quad FV = PV(1+i)^n \tag{1}$$

$$\text{Present Value (Single Sum)}: \quad PV = \frac{FV}{(1+i)^n} \tag{2}$$

$$\text{Future Value (Multiple Compounding)}: \quad FV = PV\left(1+\frac{i}{m}\right)^{nm} \tag{3}$$

$$\text{Present Value of Annuity}: \quad PV_{\text{annuity}} = PMT \times \frac{1-(1+i)^{-n}}{i} \tag{4}$$

Part 1: Bond Features and Valuation

1.1 Call Provision

Definition

A **call provision** allows the issuer to refund (repay) the bond issue if interest rates decline.

- **Benefits the issuer** (can refinance at lower rates)
- **Hurts the investor** (loses future high coupon payments)
- Borrowers pay more, and lenders require more, for callable bonds
- Most bonds have a **deferred call** and a **declining call premium**

1.2 Sinking Fund

Definition

A **sinking fund** is a provision to pay off a loan over its life rather than all at maturity.

Execution Methods

Call x% at Par	**Buy Bonds in Open Market**
Used when $r_d <$ coupon rate	Used when $r_d >$ coupon rate
Bond sells at a premium	Bond sells at a discount

1.3 Other Bond Types

Bond Type	**Description**
Convertible Bond	Exchanged for common stock at holder's option
Warrant	Long-term option to buy shares at specified price
Putable Bond	Holder can sell back to company prior to maturity
Income Bond	Pays interest only when earned by the firm
Indexed Bond	Interest rate based on inflation rate

1.4 Opportunity Cost of Debt Capital

$$\boxed{r_i = r^* + IP + MRP + DRP + LP}$$

Part 2: Bond Valuation Examples (Annual Coupons)

Example 1: Par Bond Valuation

Problem Statement: What is the value of a 10-year, 10% annual coupon bond with a \$1,000 par value, if the yield to maturity (r_d) is 10%?

Step 1: Identify the known variables.

$$\begin{aligned}
FV &= \$1,000 \\
\text{Coupon Rate} &= 10\% = 0.10 \\
PMT &= \$1,000 \times 0.10 = \$100 \\
N &= 10 \text{ years} \\
r_d &= 10\% = 0.10
\end{aligned}$$

Step 2: Write the bond valuation equation.

$$P = PMT \times \frac{1 - (1 + r_d)^{-N}}{r_d} + \frac{FV}{(1 + r_d)^N}$$

Step 3: Calculate the present value of the annuity (coupon payments).

$$\begin{aligned}
PV_{\text{annuity}} &= 100 \times \frac{1 - (1.10)^{-10}}{0.10} \\
&= 100 \times \frac{1 - 0.385543}{0.10} \\
&= 100 \times \frac{0.614457}{0.10} \\
&= 100 \times 6.14457 = \$614.46
\end{aligned}$$

Step 4: Calculate the present value of the face value.

$$PV_{\text{lump sum}} = 1,000 \times (1.10)^{-10} = 1,000 \times 0.385543 = \$385.54$$

Step 5: Sum both present values.

$$P = \$614.46 + \$385.54 = \$1,000.00$$

Answer: The bond value is $\boxed{\$1,000}$. This is a **par bond** because coupon rate = YTM.

Example 2: Premium Bond Valuation

Problem Statement: What is the value of a 10-year bond with a 13% annual coupon, $1,000 par value, and the same risk as the previous bond (YTM = 10%)?

Step 1: Identify the known variables.

$$\begin{aligned} FV &= \$1,000 \\ \text{Coupon Rate} &= 13\% = 0.13 \\ PMT &= \$1,000 \times 0.13 = \$130 \\ N &= 10 \text{ years} \\ r_d &= 10\% = 0.10 \end{aligned}$$

Step 2: Calculate the present value of the annuity.

$$\begin{aligned} PV_{\text{annuity}} &= 130 \times \frac{1-(1.10)^{-10}}{0.10} \\ &= 130 \times 6.14457 = \$798.79 \end{aligned}$$

Step 3: Calculate the present value of the face value.

$$PV_{\text{lump sum}} = 1,000 \times (1.10)^{-10} = \$385.54$$

Step 4: Sum both present values.

$$P = \$798.79 + \$385.54 = \$1,184.33$$

Answer: The bond value is $\boxed{\$1,184.33}$. This is a **premium bond** because coupon rate (13%) > YTM (10%).

Example 3: Discount Bond Valuation

Problem Statement: What is the value of a 10-year bond with a 7% annual coupon, $1,000 par value, and YTM = 10%?

Step 1: Identify the known variables.

$$\begin{aligned} FV &= \$1,000 \\ \text{Coupon Rate} &= 7\% = 0.07 \\ PMT &= \$1,000 \times 0.07 = \$70 \\ N &= 10 \text{ years} \\ r_d &= 10\% = 0.10 \end{aligned}$$

Step 2: Calculate the present value of the annuity.

$$\begin{aligned} PV_{\text{annuity}} &= 70 \times \frac{1-(1.10)^{-10}}{0.10} \\ &= 70 \times 6.14457 = \$430.12 \end{aligned}$$

Step 3: Calculate the present value of the face value.

$$PV_{\text{lump sum}} = 1,000 \times (1.10)^{-10} = \$385.54$$

Step 4: Sum both present values.

$$P = \$430.12 + \$385.54 = \$815.66$$

Answer: The bond value is $\boxed{\$815.66}$. This is a **discount bond** because coupon rate (7%) < YTM (10%).

Part 3: Yield to Maturity (YTM) Calculations

Example 4: Finding YTM for a Discount Bond

Problem Statement: What is the YTM on a 10-year, 9% annual coupon, $1,000 par value bond selling for $887?

Step 1: Identify the known variables.

$$\begin{aligned} FV &= \$1,000 \\ PMT &= \$1,000 \times 0.09 = \$90 \\ N &= 10 \\ P &= \$887 \end{aligned}$$

Step 2: Write the YTM equation (need to solve for r_d).

$$887 = 90 \times \frac{1-(1+r_d)^{-10}}{r_d} + \frac{1,000}{(1+r_d)^{10}}$$

Step 3: Trial and error - Try $r_d = 11\% = 0.11$.

$$\begin{aligned} PV_{\text{annuity}} &= 90 \times \frac{1-(1.11)^{-10}}{0.11} \\ &= 90 \times \frac{1-0.352184}{0.11} = 90 \times 5.88924 = \$530.03 \\ PV_{\text{lump sum}} &= 1,000 \times (1.11)^{-10} = 1,000 \times 0.352184 = \$352.18 \\ P_{\text{calc}} &= 530.03 + 352.18 = \$882.21 \quad \text{(Slightly below \$887)} \end{aligned}$$

Step 4: Try $r_d = 10.91\%$.

$$P_{\text{calc}} = \$887.00$$

Step 5: Use the YTM approximation formula to verify.

$$YTM \approx \frac{PMT + \frac{FV-P}{N}}{\frac{FV+P}{2}} = \frac{90 + \frac{1,000-887}{10}}{\frac{1,000+887}{2}} = \frac{90+11.3}{943.5} = \frac{101.3}{943.5} = 10.74\%$$

Answer: The YTM is $\boxed{10.91\%}$. Since YTM (10.91%) > coupon rate (9%), the bond sells at a discount.

Example 5: Finding YTM for a Premium Bond

Problem Statement: What is the YTM on a 10-year, 9% annual coupon, $1,000 par value bond selling for $1,134.20?

Step 1: Identify the known variables.

$$\begin{aligned} FV &= \$1,000 \\ PMT &= \$90 \\ N &= 10 \\ P &= \$1,134.20 \end{aligned}$$

Step 2: Trial and error - Try $r_d = 7.08\%$.

$$P_{\text{calc}} = \$1,134.20$$

Answer: The YTM is $\boxed{7.08\%}$. Since YTM (7.08%) < coupon rate (9%), the bond sells at a premium.

Part 4: Current Yield and Capital Gains Yield

Example 6: Calculating Current Yield and CGY

Problem Statement: Find the current yield and capital gains yield for a 10-year, 9% annual coupon bond selling for $887 with a $1,000 face value.

Step 1: Calculate the current yield.

$$\text{Current Yield} = \frac{\text{Annual Coupon Payment}}{\text{Current Price}} = \frac{\$90}{\$887} = 0.1015 = 10.15\%$$

Step 2: Recall the relationship between YTM, CY, and CGY.

$$YTM = \text{Current Yield} + \text{Capital Gains Yield}$$

Step 3: Solve for capital gains yield.

$$CGY = YTM - CY = 10.91\% - 10.15\% = 0.76\%$$

Answer:

$$\begin{aligned} \text{Current Yield} &= \boxed{10.15\%} \\ \text{Capital Gains Yield} &= \boxed{0.76\%} \end{aligned}$$

Part 5: Bond Values Over Time

Key Principle

At maturity, the value of any bond must equal its par value ($1,000). If r_d remains constant:

- Premium bond value **decreases** over time to $1,000
- Discount bond value **increases** over time to $1,000
- Par bond value **stays** at $1,000

Part 6: Semiannual Bonds

Adjustment Rules

1. Multiply years by 2: $N_{\text{semi}} = N \times 2$
2. Divide nominal rate by 2: $r_{\text{semi}} = r_d/2$
3. Divide annual coupon by 2: $PMT_{\text{semi}} = \text{Annual PMT}/2$

Example 7: Valuing a Semiannual Bond

Problem Statement: What is the value of a 10-year, 10% semiannual coupon bond with $1,000 par value, if $r_d = 13\%$?

Step 1: Adjust all variables for semiannual compounding.

$$\begin{aligned} FV &= \$1,000 \\ N_{\text{semi}} &= 10 \times 2 = 20 \\ r_{\text{semi}} &= 13\%/2 = 6.5\% = 0.065 \\ PMT_{\text{semi}} &= \$100/2 = \$50 \end{aligned}$$

Step 2: Calculate the present value of the annuity.

$$\begin{aligned} PV_{\text{annuity}} &= 50 \times \frac{1-(1.065)^{-20}}{0.065} \\ &= 50 \times \frac{1-0.283797}{0.065} \\ &= 50 \times \frac{0.716203}{0.065} \\ &= 50 \times 11.0185 = \$550.93 \end{aligned}$$

Step 3: Calculate the present value of the face value.

$$PV_{\text{lump sum}} = 1,000 \times (1.065)^{-20} = 1,000 \times 0.283797 = \$283.80$$

Step 4: Sum both present values.

$$P = \$550.93 + \$283.80 = \$834.73$$

Answer: The semiannual bond value is $\boxed{\$834.73}$.

Example 8: Effective Annual Rate (EAR)

Problem Statement: For the 10-year, 10% semiannual bond with $r_d = 10\%$, calculate the effective annual rate.

Step 1: Identify the nominal rate and compounding frequency.

$$\begin{aligned} r_{\text{nominal}} &= 10\% = 0.10 \\ m &= 2 \text{ (semiannual)} \end{aligned}$$

Step 2: Apply the EAR formula.

$$EAR = \left(1 + \frac{r_{\text{nominal}}}{m}\right)^m - 1 = \left(1 + \frac{0.10}{2}\right)^2 - 1 = (1.05)^2 - 1 = 1.1025 - 1 = 0.1025 = 10.25\%$$

Answer: The effective annual rate is $\boxed{10.25\%}$.

Part 7: Yield to Call (YTC)

Example 9: Finding Yield to Call

Problem Statement: A 10-year, 10% semiannual coupon bond is selling for $1,135.90. It can be called in 4 years for $1,050. What is its yield to call (YTC)?

Step 1: Identify the known variables.

$$\begin{aligned} FV &= \$1,000 \quad \text{(Not used directly for YTC)} \\ \text{Call Price} &= \$1,050 \\ PMT_{\text{semi}} &= \$100/2 = \$50 \\ N_{\text{call}} &= 4 \times 2 = 8 \text{ semiannual periods} \\ P &= \$1,135.90 \end{aligned}$$

Step 2: Write the YTC equation.

$$1,135.90 = 50 \times \frac{1-(1+r_{\text{call}})^{-8}}{r_{\text{call}}} + \frac{1,050}{(1+r_{\text{call}})^8}$$

Step 3: Solve for the periodic YTC.

$$r_{\text{call, periodic}} = 3.568\%$$

Step 4: Convert to nominal annual YTC.

$$YTC_{\text{nominal}} = 3.568\% \times 2 = 7.137\%$$

Step 5: Calculate the effective YTC.

$$YTC_{\text{effective}} = (1.03568)^2 - 1 = 1.0726 - 1 = 7.26\%$$

Answer:

$$\begin{aligned} \text{Periodic YTC} &= \boxed{3.568\%} \\ \text{Nominal YTC} &= \boxed{7.137\%} \\ \text{Effective YTC} &= \boxed{7.26\%} \end{aligned}$$

Example 10: YTM vs. YTC Decision

Problem Statement: A callable bond has a 10% coupon rate, YTC = 7.137%, and YTM = 8%. Which return are investors more likely to earn?

Step 1: Compare the coupon rate to current market rates.

- Coupon rate = 10%
- YTC = 7.137%
- The firm can issue new bonds at 7.137%

Step 2: Calculate potential interest savings.

$$\text{Savings per bond} = \$100 - \$71.37 = \$28.63 \text{ per year}$$

Step 3: Determine the likely outcome. Since the firm can save money by calling the bond and refinancing at a lower rate, investors should expect a call.

Answer: Investors should expect to earn the $\boxed{YTC \text{ of } 7.137\%}$ rather than the YTM of 8%.

When is a Call More Likely?

If a bond sells at a **premium** (coupon rate > market rate), a call is more likely.

- Expect to earn **YTC** on premium bonds
- Expect to earn **YTM** on par and discount bonds

Part 8: Price Risk vs. Reinvestment Risk

Price Risk

Definition

Price risk is the concern that rising r_d will cause the value of a bond to fall.

Example 11: Comparing Price Risk

Problem Statement: Which bond has more price risk: a 1-year bond or a 10-year bond? Assume both have 10% annual coupons and $r_d = 10\%$.

Step 1: Calculate the 1-year bond price at different rates.

$$\text{At } r = 10\%: \quad P = \$1,000$$
$$\text{At } r = 15\%: \quad P = \frac{\$100 + \$1,000}{1.15} = \$956.52$$
$$\text{Price decline} = \frac{956.52 - 1,000}{1,000} = -4.35\%$$

Step 2: Calculate the 10-year bond price at different rates.

$$\text{At } r = 10\%: \quad P = \$1,000$$
$$\text{At } r = 15\%: \quad P = 100 \times \frac{1 - (1.15)^{-10}}{0.15} + 1,000 \times (1.15)^{-10} = \$707.63$$
$$\text{Price decline} = \frac{707.63 - 1,000}{1,000} = -29.24\%$$

Answer: The **10-year bond** has more price risk (29.24% decline vs. 4.35%).

Reinvestment Risk

Definition

Reinvestment risk is the concern that r_d will fall, and future cash flows will have to be reinvested at lower rates, reducing income.

Example 12: Reinvestment Risk Scenario

Problem Statement: You win $500,000 in the lottery. You can invest in either a 10-year bond or a series of ten 1-year bonds. Both currently yield 10%. If 1-year rates fall to 3% after one year, what happens to your annual income under each strategy?

Step 1: Analyze the 1-year bond strategy.

$$\text{Year 1 income} = \$500,000 \times 10\% = \$50,000$$
$$\text{After Year 1, reinvest at 3\%}: \quad \text{New annual income} = \$500,000 \times 3\% = \$15,000$$
$$\text{Income drop} = \$50,000 - \$15,000 = \$35,000$$

Step 2: Analyze the 10-year bond strategy.

$$\text{Annual income locked in} = \$500,000 \times 10\% = \$50,000 \text{ for all 10 years}$$

Answer: The 1-year bond strategy exposes you to significant **reinvestment risk**. The 10-year bond strategy locks in the higher rate.

Summary: Price Risk vs. Reinvestment Risk

Bond Type	Price Risk	Reinvestment Risk
Short-term / High-coupon bonds	Low	High
Long-term / Low-coupon bonds	High	Low

Conclusion

NOTHING IS RISKLESS!

Part 9: Default Risk and Bond Ratings

Default Risk

Definition

If an issuer defaults, investors receive less than the promised return. Default risk is influenced by the issuer's financial strength and the terms of the bond contract.

Bond Rating Categories

Rating	Moody's	S&P
Investment Grade		
Highest Quality	Aaa	AAA
High Quality	Aa	AA
Upper Medium	A	A
Medium Grade	Baa	BBB
Junk Bonds		
Speculative	Ba	BB
Highly Speculative	B	B
Substantial Risk	Caa	CCC

Note

Bond ratings are designed to reflect the **probability of default**.

Factors Affecting Bond Ratings

1. **Financial Performance:** Debt ratio, TIE ratio, current ratio
2. **Qualitative Factors:** Secured vs. unsecured debt, senior vs. subordinated debt
3. **Other Factors:** Earnings stability, regulatory environment, pension liabilities

Part 10: Bankruptcy

Two Main Chapters of the Federal Bankruptcy Act

Chapter 11: Reorganization	Chapter 7: Liquidation
Business continues to operate	Business ceases operations
Management usually stays in control	Assets are auctioned off
Company has 120 days to file a plan	Proceeds distributed by priority
Firm must be "worth more alive than dead"	Used if worth more "dead than alive"

Priority of Claims in Liquidation (Chapter 7)

1. Secured creditors (from sale of secured assets)
2. Trustee's costs
3. Wages (subject to limits)
4. Taxes
5. Unfunded pension liabilities
6. Unsecured creditors
7. Preferred stockholders
8. Common stockholders

Key Insight

In liquidation, unsecured creditors generally receive **nothing**. This makes them more willing to participate in reorganization even though their claims are greatly scaled back.

Reorganization Process

1. Various groups of creditors vote on the reorganization plan
2. If majority of creditors AND the judge approve:
3. The company "emerges" from bankruptcy with:
   - Lower debts

- Reduced interest charges
- A chance for success

Formula Summary Sheet

All Bond Valuation Formulas

$$\text{Bond Price (Annual)}: \quad P = PMT \times \frac{1-(1+r_d)^{-N}}{r_d} + \frac{FV}{(1+r_d)^N} \tag{5}$$

$$\text{Bond Price (Semiannual)}: \quad P = \frac{PMT}{2} \times \frac{1-(1+r_d/2)^{-2N}}{r_d/2} + \frac{FV}{(1+r_d/2)^{2N}} \tag{6}$$

$$\text{YTM Approximation}: \quad YTM \approx \frac{PMT + \frac{FV-P}{N}}{\frac{FV+P}{2}} \tag{7}$$

$$\text{Current Yield}: \quad CY = \frac{PMT}{P} \tag{8}$$

$$\text{Capital Gains Yield}: \quad CGY = YTM - CY \tag{9}$$

$$\text{Effective Annual Rate}: \quad EAR = \left(1 + \frac{r_{\text{nominal}}}{m}\right)^m - 1 \tag{10}$$

$$\text{Opportunity Cost}: \quad r_i = r^* + IP + MRP + DRP + LP \tag{11}$$